

DELIVERABLE 1

Due by Wednesday, September 23rd, 2026, 11:59 PM.

This is a team deliverable. Work with your assigned team members only. Each team needs to upload its solutions to Canvas as a single PDF file (any team member can do so through their Canvas account). All team members should contribute to the deliverable. If a team member did not contribute, then their name should not appear on the cover page. Scripts should be written in Python.

Problem 1 (25 points)

Cambridge Computers asked for your help in a project that seeks to better inform their pricing strategy for laptops. You have been provided a dataset with the following variables:

Variable	Definition
InventoryID	Inventory ID of laptop for internal stocking purposes
Company	Manufacturer of laptop: Asus, Dell, HP, or Lenovo
TypeName	Type of laptop: Gaming, Notebook, or Ultrabook
GPU	Provider of GPU: AMD, Intel, Nvidia
Screen	Screen size of laptop in inches
Memory	Laptop's RAM in GBs
Weight	Weight of laptop in KG
Rating	The average customer rating of the laptops from an online review site
Price	Price of laptop [in Euros]

Table 1: Variables and definitions for Cambridge Computers dataset.

You have been tasked with the job of building a good model to predict the price of a laptop using the variables provided. The training and test datasets are available on Canvas: `laptop_train.csv` and `laptop_test.csv`. In this dataset, you should treat the variables Screen, Memory, Weight, Rating, and Price as numeric variables.

Answer the following questions.

- Build a good linear statistical model to predict the price of a laptop using the data provided. Outline the procedure you followed to arrive at your final model. Provide a screenshot of the Python output from `summary()` for your final fitted regression model.
- Discuss which of the independent variables in your final model are managerially sensible to you, and which, if any, are worthy of further investigation.
- Based on your model, which laptop manufacturer has the highest effect on the price of the laptop? Which laptop manufacturer has the smallest effect?
- What is the out-of-sample R-squared of your model? Give a precise interpretation of it.
- Consider a laptop with the following characteristics:

InventoryID = 950, Company = Asus, TypeName = Ultrabook, GPU = Intel,
Screen = 15.6, Memory = 6, Weight = 3, Rating = 8.

For this laptop, what is the probability that the predicted price will exceed 1,100 Euros using your final model? Please state clearly the assumptions you are making, if any, to arrive at your answer.

In linear regression models, an interaction term or effect between two predictors x_1 and x_2 denotes the product of these two predictors, i.e., x_1x_2 . Using the `statsmodels.formula.api` module, you can fit a linear regression model for response y using x_1 , x_2 , and their interaction with:

```
model = smf.ols("y ~ x1 + x2 + x1:x2", data=your_data).fit()
```

Equivalently, the formula `"y ~ x1 * x2"` includes both main effects and their interaction.

- f) Modify your model from part (a) by including an interaction term between Memory and GPU. What is the interpretation of the coefficient corresponding to this interaction effect? Comment briefly (at most 2 sentences) on how this new model differs from the part (a) model.
- g) Modify your model from part (a) by including an interaction term between GPU and Company. What is the interpretation of the coefficient corresponding to this interaction effect? Comment briefly (at most 2 sentences) on how this new model differs from the model in part (a).

For parts (a), (f), and (g), make sure categorical variables are appropriately specified in a `statsmodels` formula using `C(variable_name)`.

Problem 2 (30 points)

Inference in statistical models is a crucial aspect of data-driven decision-making, particularly in industries such as insurance, where accurate risk assessment and pricing are essential. However, the misuse or misinterpretation of these models can lead to serious consequences, such as unfair pricing practices or misjudgment of risk exposure.

To better understand the factors affecting health insurance premiums, you have been provided with a dataset `insurance.csv` on Canvas containing information on 1,338 insured individuals. The dataset includes the following attributes for each policyholder: Age, Sex, BMI, Number of Children, Smoker status (Y/N), and Region, along with their corresponding insurance charges.

Your task is to analyze the dataset and build models to gain insights into the risk underwriting process and the relationship between policyholder attributes and insurance charges. This analysis will involve examining the distribution of insurance charges, assessing the fit of various probability distributions, and building a regression model to predict charges based on policyholder attributes.

- a) Load the insurance dataset (`insurance.csv`) as a pandas DataFrame using `pd.read_csv()`.
- b) Calculate and report the mean and sample standard deviation of the insurance `charges` variable using the pandas `mean()` and `std()` methods.
- c) Assuming the charges for the individuals follow a normal distribution with your mean and standard deviation from part (b), use `scipy.stats.norm.cdf()` to determine the probability that a randomly selected policyholder has an insurance charge between \$8,000 and \$14,000.
- d) Using the actual historical data, calculate the proportion of individuals with an insurance charge between \$8,000 and \$14,000. Does this align with the probability you obtained when assuming the charges follow a normal distribution in part (c)? Justify why or why not.
- e) The insurance company is curious to explore different probability models for insurance charges. Fit normal, lognormal, uniform, and Cauchy distributions using the corresponding `scipy.stats` distribution objects and their `fit()` methods. Compare the fitted distributions visually in two ways: (i) overlay each fitted probability density on a density-scaled histogram of the observed charges, and (ii) compare

each fitted cumulative distribution function with the empirical cumulative distribution function of the observed charges. Which distribution appears to fit best? Justify your choice by discussing discrepancies in both the center and tails of the distribution. How can you refine your conclusion from part (d) in light of this?

Hint: The relevant SciPy objects are `scipy.stats.norm`, `scipy.stats.lognorm`, `scipy.stats.uniform`, and `scipy.stats.cauchy`. If `params = distribution.fit(charges)`, evaluate the fitted density and CDF using `distribution.pdf(x, *params)` and `distribution.cdf(x, *params)`. For a density-scaled histogram, use `plt.hist(charges, density=True)`.

- f) Using `statsmodels.formula.api.ols()`, build an appropriate regression model to predict insurance charges using all of the given attributes of the insured (no variable selection and no variable transformation are needed). Provide the regression output returned by the fitted model's `summary()` method.

Hint: In a `statsmodels` formula, categorical variables can be specified using `C(variable_name)`.

- g) Interpret the regression coefficients of your model from part (f) in order to explain the insurance company's pricing strategy to a regulator. In particular, explain how the company is setting its charges as a function of each independent variable included in the model. For every variable, interpret the coefficient's sign and magnitude, as well as its statistical significance.
- h) Due to new regulations, age is now considered a sensitive attribute and cannot be used in any predictive modeling. Refit your model from part (f) with this variable omitted. How does this change affect the model's coefficient of determination (R^2) evaluated on the training dataset? Comment on whether you think some age groups are more likely to be undercharged or overcharged when age is not considered as an independent variable.
- i) In light of your findings in part (e), would you recommend transforming the response variable for part (f)? If yes, outline in one sentence how you could redo your analysis in part (f). (No need to carry out the full analysis here.)

Problem 3 (25 points)

In this problem, we will use the `mtcars` dataset, which contains data extracted from the 1974 *Motor Trend* US magazine and comprises fuel consumption and 10 aspects of automobile design and performance for 32 automobiles (1973–74 models). You can load the dataset in Python using:

```
import statsmodels.api as sm
mtcars = sm.datasets.get_rdataset("mtcars", "datasets").data
```

Suppose we are interested in predicting *mpg* (miles per gallon) using a linear model with the independent variables

- *hp* (horsepower),
- *wt* (weight), and
- the interaction between *hp* and *wt*.

We will denote this as Model 1.

- a) Fit Model 1 to the `mtcars` dataset using `statsmodels.formula.api.ols()` and show the fitted model's `summary()` output.
- b) In Model 1, what is the interpretation of the coefficients of the main effects *hp* and *wt*? What is the interpretation of the coefficient of the interaction effect $hp \times wt$?

Let

$$hp_c = hp - \overline{hp} \quad \text{and} \quad wt_c = wt - \overline{wt}$$

denote the mean-centered versions of the predictors hp and wt , respectively.

- c) You now consider Model 2, which fits response mpg using the independent variables hp_c , wt_c , and their interaction $hp_c wt_c$. Create the centered columns using pandas. Fit Model 2 using `smf.ols()`, show your output, and comment on how its coefficients compare with those of Model 1.
- d) You now consider Model 3, which fits response mpg using only the interaction term $hp_c wt_c$. In other words, Model 3 includes the interaction effect but does not include the main effects hp_c and wt_c . Fit Model 3 using Python, show your output, and comment on how its coefficients compare with those of Models 1 and 2.

Problem 4 (20 points)

Let's go back to Problem 1 (Cambridge Computers). Instead of predicting the raw price, management wants to determine whether a particular computer will be high-priced (expensive) or not. They consider the price of a computer high if it is 500 Euros or higher, and low if it is 499.99 Euros or lower. Start by creating a new binary column `high` in both the test and train DataFrames, which is 1 if the price is 500 Euros or higher and 0 otherwise. Our goal is to build a logistic regression model predicting whether the price is high based on the independent variables in Table 1.

- a) Build a logistic regression model using the train dataset in which you predict whether the price is high or low using the other independent variables. Use `statsmodels.formula.api.logit()` and show the fitted model's `summary()` output. Do not perform any variable selection.

Answer questions (b)–(g) using your answer for part (a):

- b) Which variables are significant in predicting the probability of a price being high?
- c) Comment on whether the significant variables of the logistic regression model are the same as the significant variables in the linear regression model from Problem 1.
- d) For each significant variable, comment on whether an increase in its value increases or decreases the probability of a computer being high-priced. Comment on whether this makes sense.
- e) For which independent variables are the signs of the coefficients the same between the logistic regression model and the linear model? For which are they different?
- f) Consider a laptop with the following characteristics:
 - InventoryID: 4096
 - Company: Lenovo
 - TypeName: Ultrabook
 - GPU: Intel
 - Screen: 8
 - Memory: 8
 - Weight: 4.2
 - Rating: 7

Write down the equation for the probability of its being high-priced and calculate that probability.

- g) Now apply your model to the test dataset. Using a probability cutoff of 0.5, what is the accuracy of your model on the test dataset?